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MOBILE PROGRAMMING

Bagian 1:

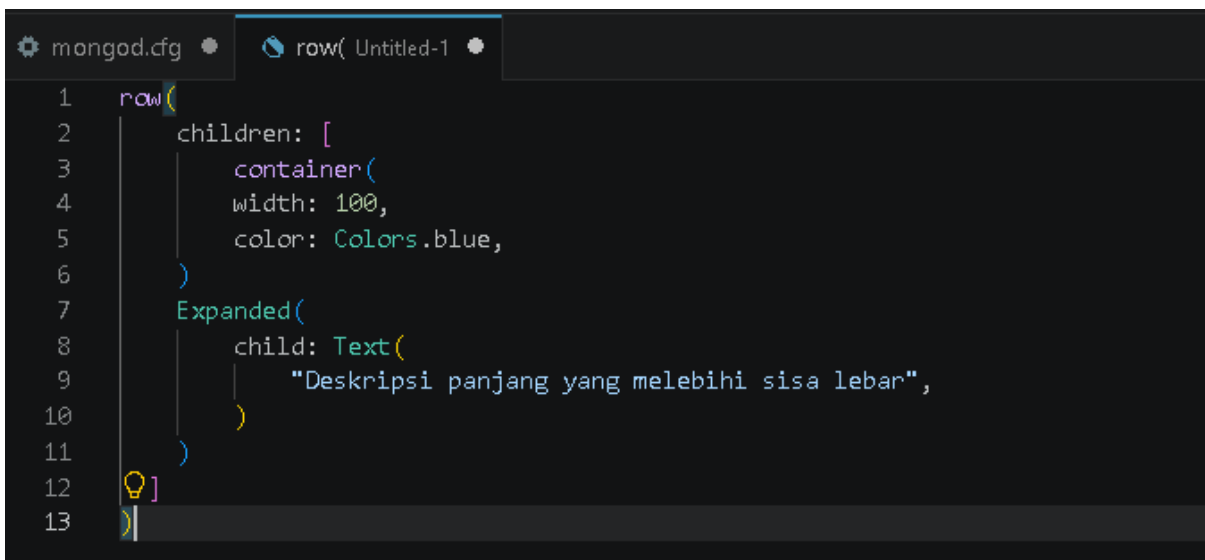
1. Perbedaan Child dan Children, Child adalah untuk menampung satu widget, seperti pada container dan center. Sedangkan Children digunakan untuk menampung beberapa widget dalam bentuk list<widget>, seperti pada Row dan Column.

2. Jika isi Row memiliki lebar yang melebihi ruang horizontal yang tersedia. Flutter dapat menampilkan overflow error berupa garis kuning dan hitam pada bagian yang melampaui batas layer, biasanya ditambahi pesan seperti:

“ a renderflex overflowed by Pixels on the right.”

Untuk membuat sebuah elemen menggunakan sisa ruang yang tersedia, gunakan widget expanded.

Contoh:



```
1 row(  
2   children: [  
3     container(  
4       width: 100,  
5       color: Colors.blue,  
6     )  
7     Expanded(  
8       child: Text(  
9         "Deskripsi panjang yang melebihi sisa lebar",  
10      )  
11   )  
12 ]  
13 )
```

3. Perbedaan MainAxisAlignment

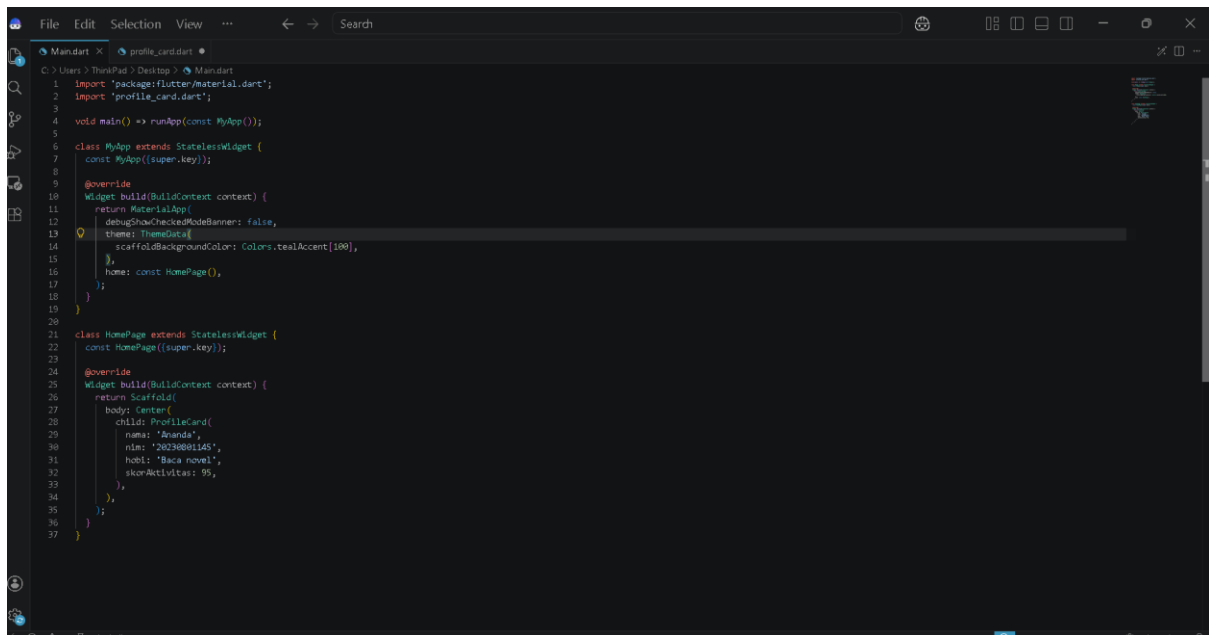
Pada column, sumbu utama adalah vertical.

MainAxisAlignment.center

Tiga widget anak akan dikelompokkan di Tengah column, sehingga ruang kosong terutama antara widget, termasuk ruang sebelum widget pertama dan setelah widget terakhir.

Bagian 2:

Kode bagian Main



```
File Edit Selection View ... Search
C:\Users\ThinkPad\Desktop> Mandart
1 import 'package:flutter/material.dart';
2 import 'profile_card.dart';
3
4 void main() => runApp(const MyApp());
5
6 class MyApp extends StatelessWidget {
7   const MyApp({super.key});
8
9   @override
10  Widget build(BuildContext context) {
11    return MaterialApp(
12      debugShowCheckedBanner: false,
13      theme: ThemeData(
14        scaffoldBackgroundColor: Colors.tealAccent[100],
15      ),
16      home: const HomePage(),
17    );
18  }
19 }
20
21 class HomePage extends StatelessWidget {
22   const HomePage({super.key});
23
24   @override
25  Widget build(BuildContext context) {
26    return Scaffold(
27      body: Center(
28        child: ProfileCard(
29          nama: 'Ananda',
30          nrm: '20230801145',
31          hobi: 'Baca novel',
32          skorAktivitas: 95,
33        ),
34      ),
35    );
36  }
37 }
```

Kode Bagian Profile_card

```
File Edit Selection View ... Search
Maindart profile_card.dart
C:\Users> ThonPad > Desktop > profile_card.dart
1 import 'package:flutter/material.dart';
2
3 class ProfileCard extends StatelessWidget {
4   final String nama;
5   final String nim;
6   final String hobi;
7   final int skorAktivitas;
8
9   const ProfileCard({
10     super.key,
11     required this.nama,
12     required this.nim,
13     required this.hobi,
14     required this.skorAktivitas,
15   });
16
17   @override
18   Widget build(BuildContext context) {
19     // Perhitungan berdasarkan NIM 20240801145
20     const double lebarKartu = 320.0 * (4 * 5);
21     const double radiusKartu = 12.0 * (5 * 1.5);
22     const double ukuranLogo = 60.0 * (5 * 2);
23     const double jarakPemisah = 15.0 * 5;
24
25     return Container(
26       width: lebarKartu,
27       padding: const EdgeInsets.all(16.0),
28       decoration: BoxDecoration(
29         color: Colors.white,
30         borderRadius: BorderRadius.circular(radiusKartu),
31         boxShadow: [
32           BoxShadow(
33             color: Colors.black.withOpacity(0.2),
34             blurRadius: 16.0,
35           ),
36         ],
37       ),
38       child: Column(
39         mainAxisAlignment: MainAxisAlignment.min,
40         children: [
41           Row(
42             children: [
43               Container(
44                 padding: const EdgeInsets.all(8.0),
```

```
File Edit Selection View ... Search
Maindart profile_card.dart
C:\Users> ThonPad > Desktop > profile_card.dart
1 class ProfileCard extends StatelessWidget {
2   Widget build(BuildContext context) {
3     return Container(
4       width: 320.0 * (4 * 5),
5       padding: const EdgeInsets.all(16.0),
6       decoration: BoxDecoration(
7         color: Colors.white,
8         borderRadius: BorderRadius.circular(12.0 * (5 * 1.5)),
9         boxShadow: [
10           BoxShadow(
11             color: Colors.black.withOpacity(0.2),
12             blurRadius: 16.0,
13           ),
14         ],
15       ),
16       child: Column(
17         mainAxisAlignment: MainAxisAlignment.min,
18         children: [
19           Row(
20             children: [
21               Container(
22                 padding: const EdgeInsets.all(8.0),
23                 child: Column(
24                   mainAxisAlignment: MainAxisAlignment.start,
25                   children: [
26                     Text(
27                       'Nama: $nama',
28                       style: TextStyle(
29                         fontSize: 16,
30                         fontWeight: FontWeight.bold,
31                       ),
32                     ),
33                     Text(
34                       'nim',
35                       style: TextStyle(
36                         fontSize: 16,
37                       ),
38                     ),
39                     const Divider(
40                       thickness: 1.0,
41                     ),
42                     Column(
43                       mainAxisAlignment: MainAxisAlignment.start,
44                       children: [
45                         Text(
46                           'Nama: $nama',
47                           style: TextStyle(
48                             fontSize: 16,
49                             fontWeight: FontWeight.normal,
50                           ),
51                         ),
52                         Text(
53                           'nim: $nim',
54                           style: TextStyle(
55                             fontSize: 16,
56                             fontWeight: FontWeight.normal,
57                           ),
58                         ),
59                         Text(
60                           'Hobi: $hobi',
61                           style: TextStyle(
62                             fontSize: 16,
63                             fontWeight: FontWeight.normal,
64                           ),
65                         ),
66                         Text(
67                           'Skor Aktivitas: $skorAktivitas',
68                           style: TextStyle(
69                             fontSize: 16,
70                             fontWeight: FontWeight.normal,
71                           ),
72                         ),
73                       ],
74                     ),
75                   ],
76                 ),
77             ],
78           ),
79         ],
80       ),
81     );
82   }
83 }
```

Note: IBM_BOB